Neural Network Architecture

Course	**Neural Network Architecture.pptx**
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Designed by	**Upskills**
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Objectives

- Learn neural network architectures and apply them for regression and classification tasks using TensorFlow.

1. Introduction to Neural Network Architecture

- Neural networks are computing systems inspired by the biological neural networks that constitute animal brains.
- It consists of interconnected groups of artificial neurons.
- It recognizes patterns, make decisions, and predict outcomes.

1. Introduction to Neural Network Architecture

- Key Components of Neural Networks
  - Layers:
    - Input Layer: Takes in the data.
    - Hidden Layers: Process the data through weights and activation functions.
    - Output Layer: Produces the final result.
  - Neurons: Basic units that process inputs and produce outputs.
  - Weights and Biases: Parameters that the model learns during training.
  - Activation Functions: Introduce non-linearity to the model.

1. Introduction to Neural Network Architecture

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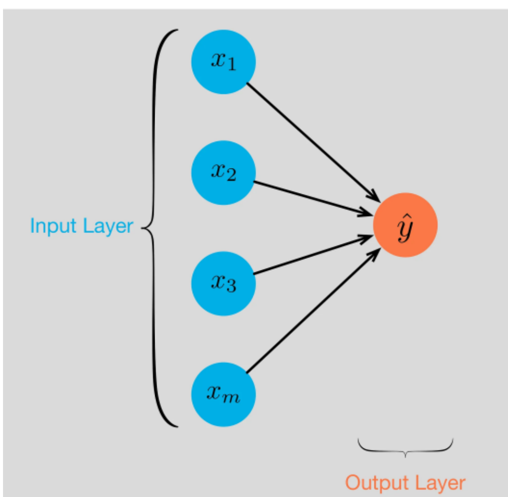
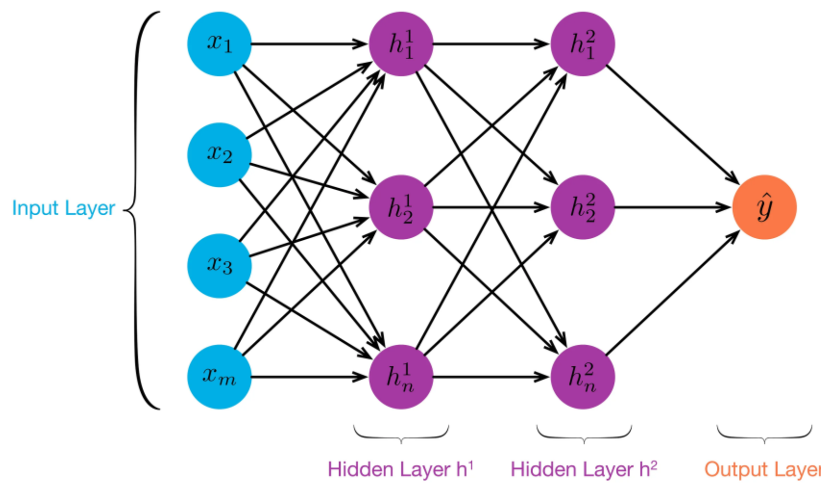
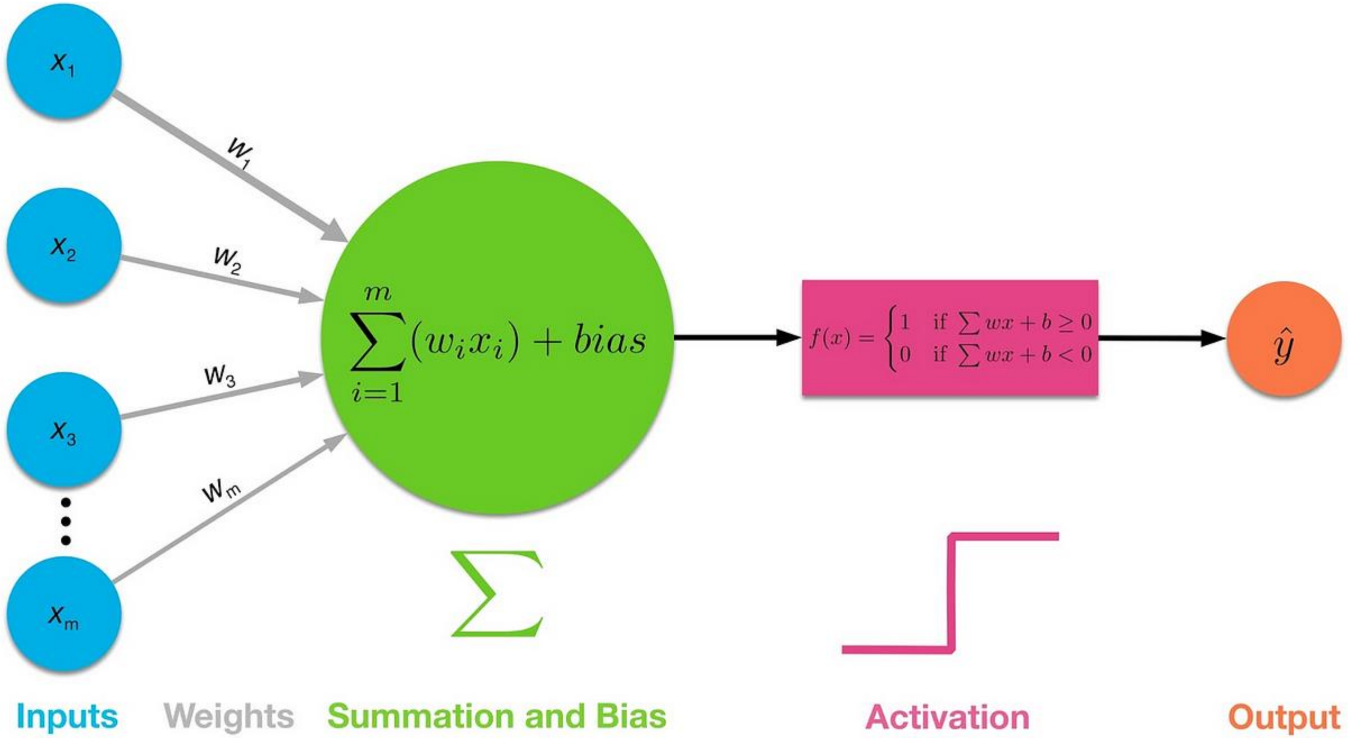




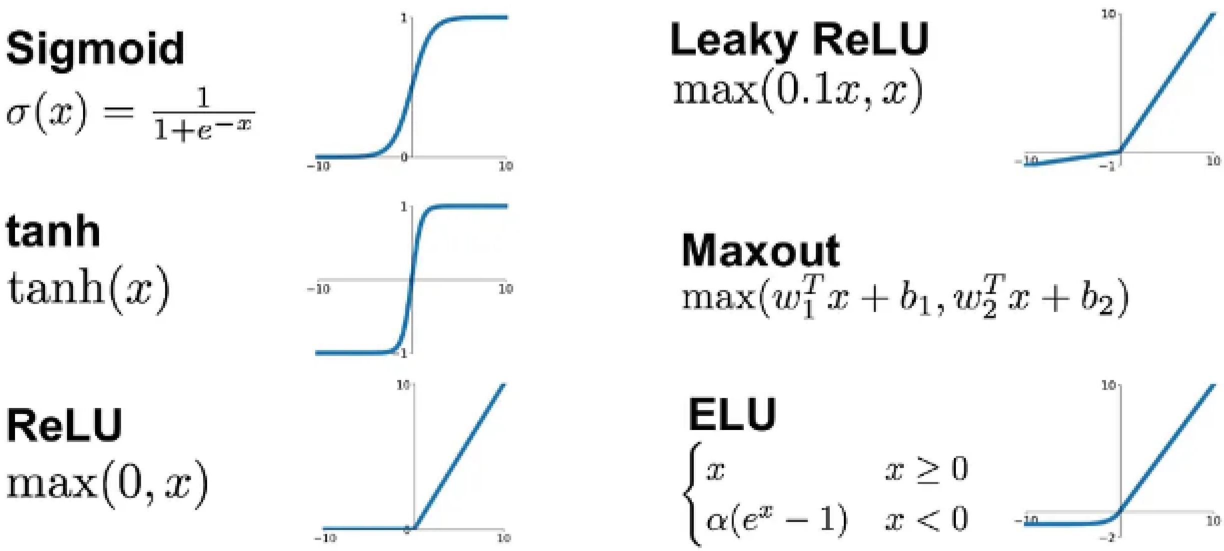
1. Introduction to Neural Network Architecture

- Neurons:
  - Receive inputs, apply weights, sum them up, add a bias, and pass through an activation function.



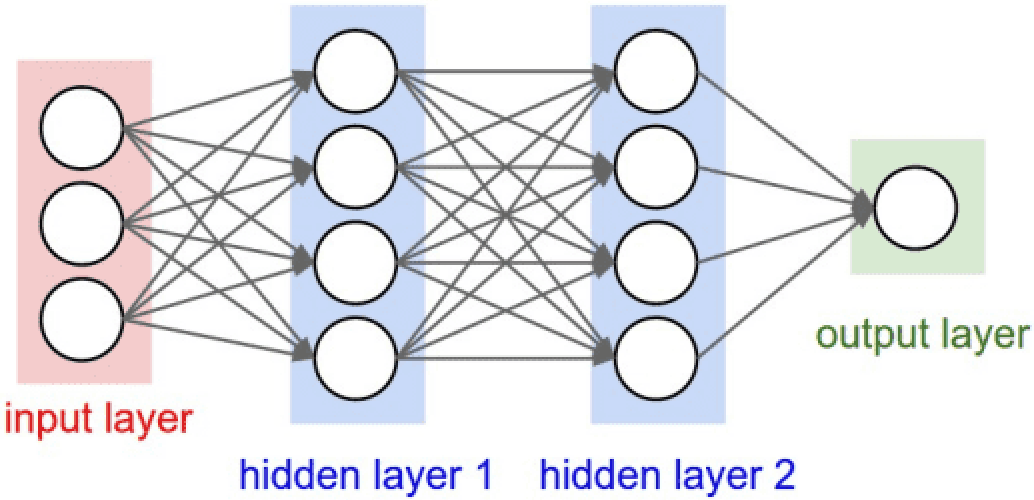
1. Introduction to Neural Network Architecture

- Activation Functions



2. Feed Forward Neural Networks

- Feed Forward Neural Networks
  - A type of artificial neural network where connections between the nodes do not form a cycle.
  - Data flows in one direction, from input to output

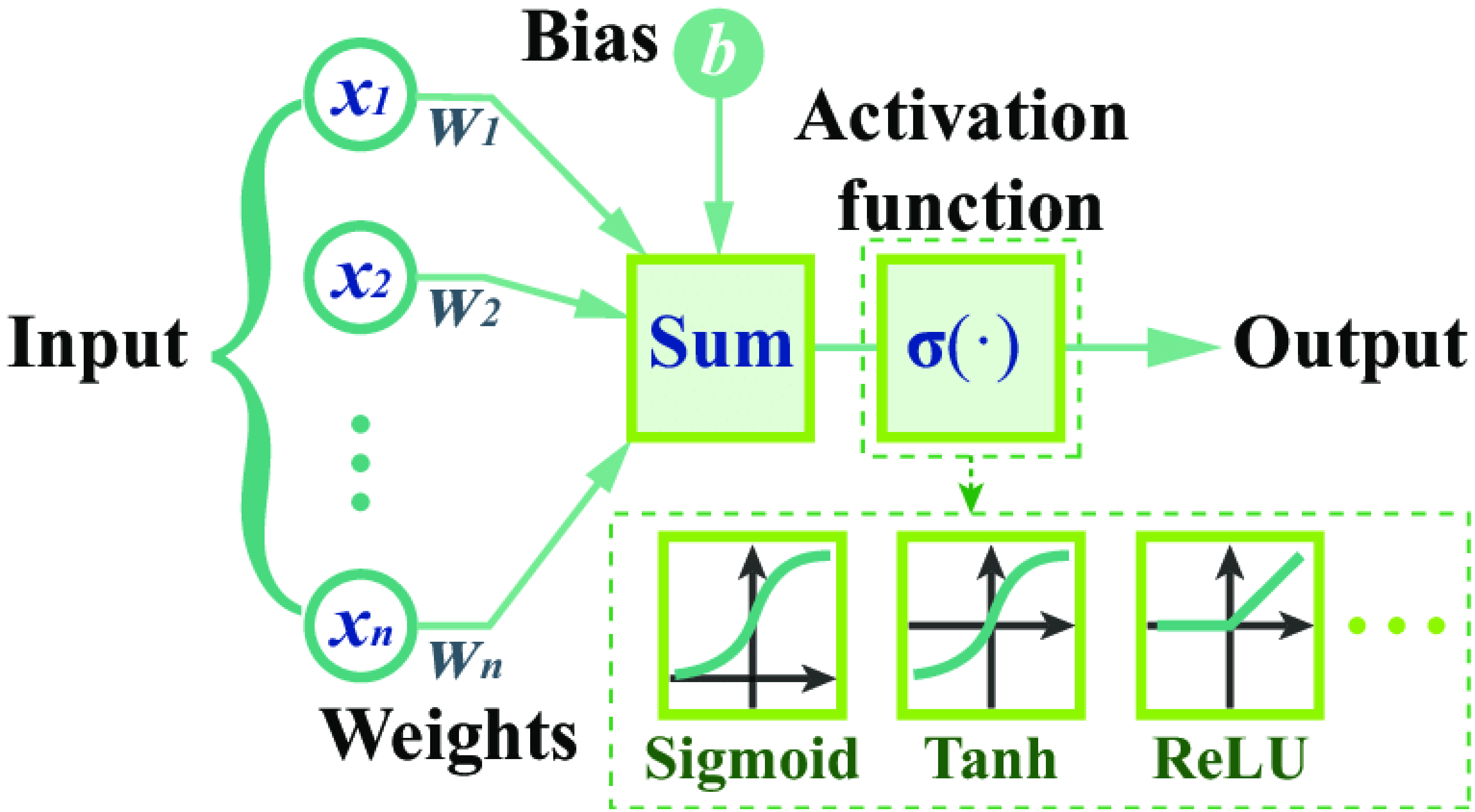


2.Feed Forward Neural Network - Structure

- Feed Forward Neural Networks
  - Layers:
    - Input Layer: Neurons that take the input features.
    - Hidden Layers: One or more layers of neurons that apply transformations.
    - Output Layer: Neurons that produce the final output.
  - Neurons: Basic units that perform computations.
  - Weights: Parameters that transform the input data within the neurons.
  - Biases: Additional parameters in neurons to adjust the output.

2. Feed Forward Neural Networks

- Feed Forward Neural Networks
  - Forward Propagation
    - Definition: Process of moving inputs through the network to



2. Feed Forward Neural Networks

- Feed Forward Neural Networks
  - Backpropagation